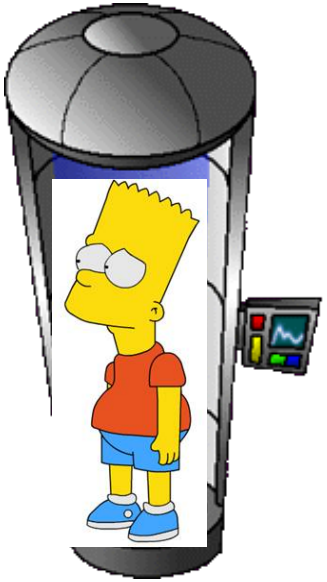
CSEP 590tv: Quantum Computing

Dave Bacon

July 20, 2005



Today's Menu

Administrivia

Finish Teleportation

Superdense Coding

n Qubit registers

Begin Quantum Algorithms



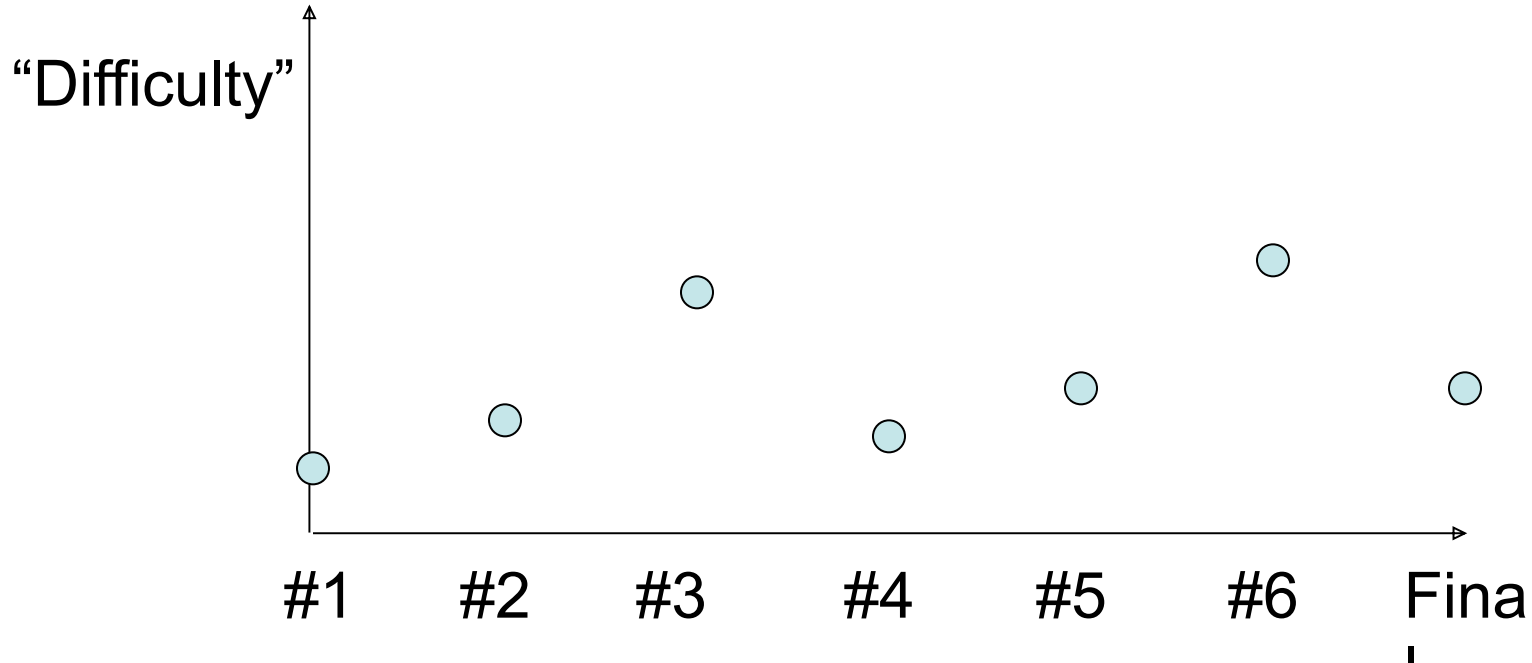
Administrivia

Turn in HW #3. It was meant to be harder than HWs #1 and #2. Was it?

Pick up HW #4. It should be easier than HW #3.

HW#2 solutions available on website

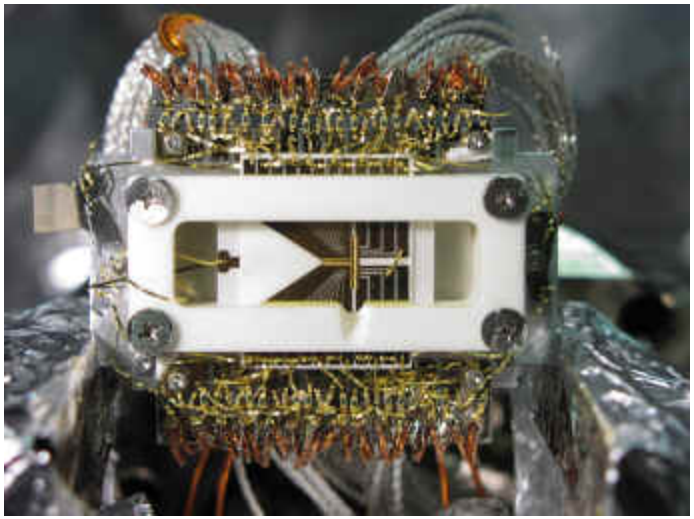
Here is my goal:

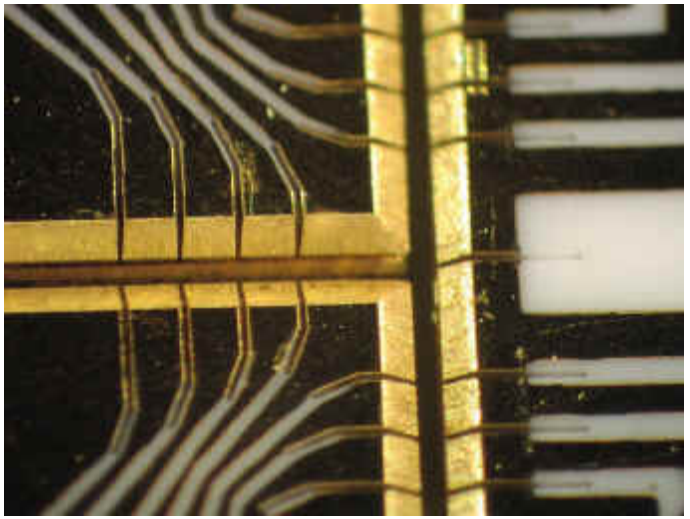


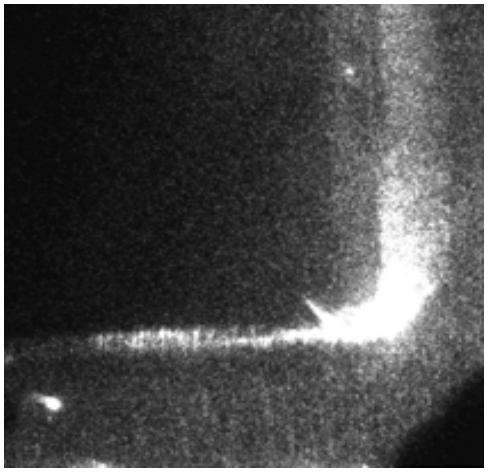
Administrivia

July 20: multi qubit registers, begin quantum algorithms
July 27: quantum algorithms
Aug 3: quantum entanglement
Aug 10: quantum error correction
Aug 17: post-final lecture on ???

Shuttling Around a Corner







Pictures snatched from Chris Monroe's University of Michigan website

Reca

p

Matrices in outer product notati

$$M = \sum_{i,j=0}^{N-1} M_{i,j}|i\rangle\langle j| = \begin{bmatrix} M_{0,0} & \cdots & M_{N-1,0} \\ \vdots & \ddots & \vdots \\ M_{N-1,0} & \cdots & M_{N-1,N-1} \end{bmatrix}$$

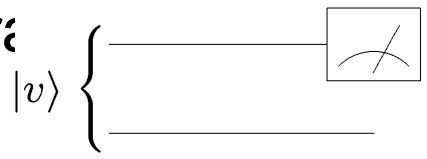
Projectors: $P = |v\rangle\langle v|$

Measurement operators: $M_i = |v_i\rangle\langle v_i|$

Probability of outcome i: $Pr(i) = \langle v|M_i|v\rangle$

New state if outcome is i: $|v'\rangle = \dfrac{M_i|v\rangle}{\sqrt{Pr(i)}}$

Measuring first of two qubits. Measurement opera



$$Pr(0) = \langle v|M_0 \otimes I|v\rangle \qquad M_0 \otimes I$$

$$Pr(1) = \langle v|M_1 \otimes I|v\rangle \qquad M_1 \otimes I$$

Recap

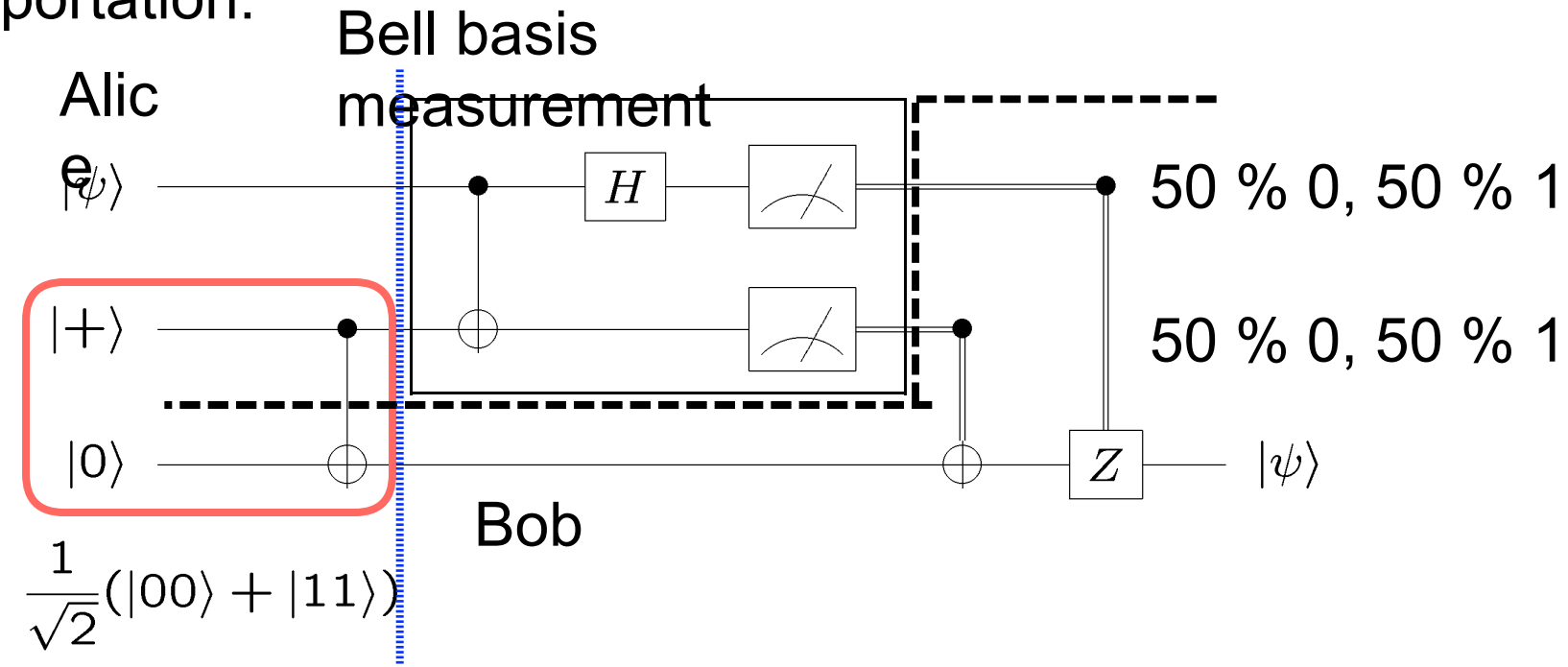
Deutsch's problem.

Distinguishing between constant and balanced.

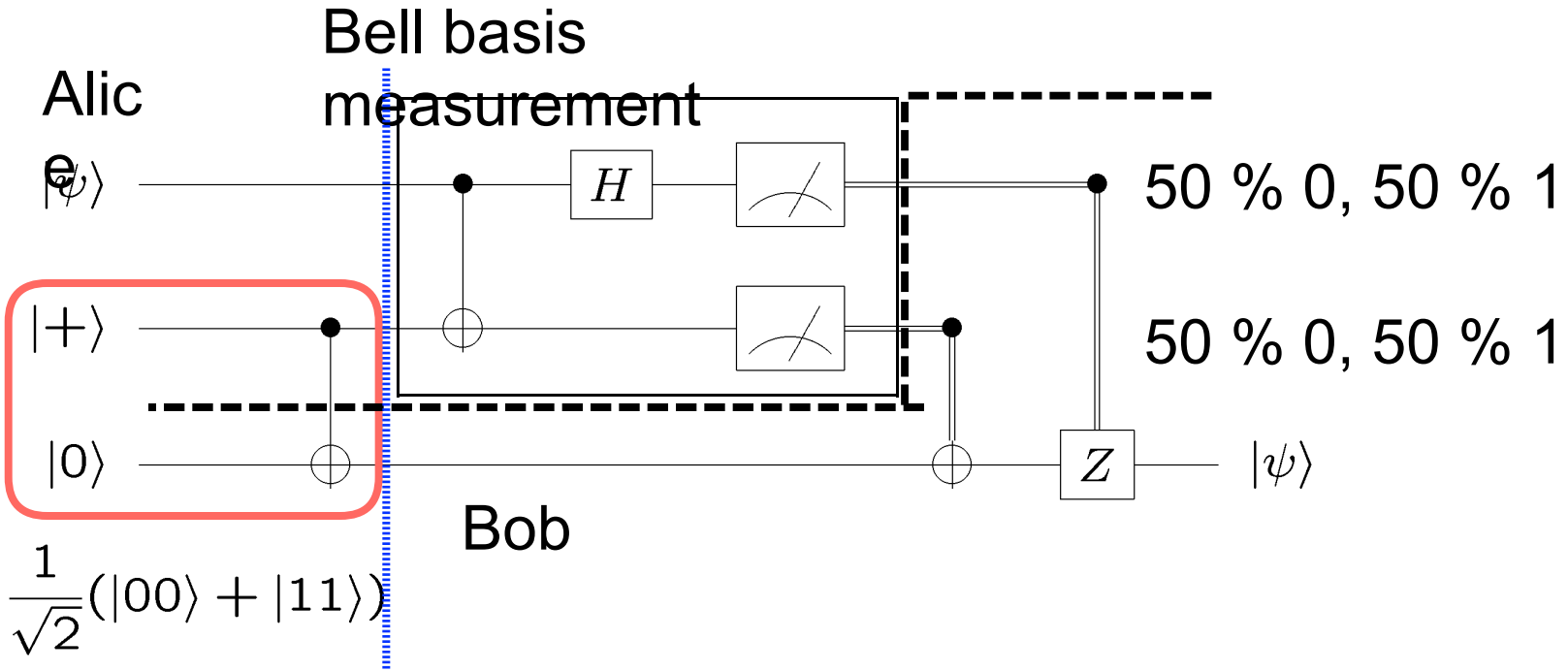
2 classical queries

1 quantum query

Quantum teleportation:



Teleportation



First step is that Alice and Bob should share the entangled state:

$$|+\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$$

$|0\rangle$

$$\frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$$

Teleportation

two qubits



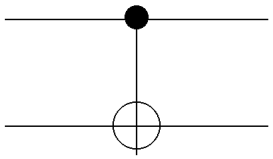
ALICE



BOB

2.
Separate
1. Interact and entangle

$$|+\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$$



$$|0\rangle$$

$$\frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$$

Alice and Bob each have a qubit, and the wave function of their two qubit is entangled. This means that we can't think of Alice's qubit as having a particular wave function. We have to talk about the "global" two qubit wave function.

$$\frac{1}{\sqrt{2}}(|00\rangle + |11\rangle) \neq |a\rangle \otimes |b\rangle$$

Teleportation



ALICE

$$\frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$$



BOB

Alice does not know the wave function

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$$

We have three qubits whose wave function is

$$|\psi\rangle \otimes \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$$

qubit 1 — qubit 2 and qubit 3

Separable, Entangled, 3 Qubits

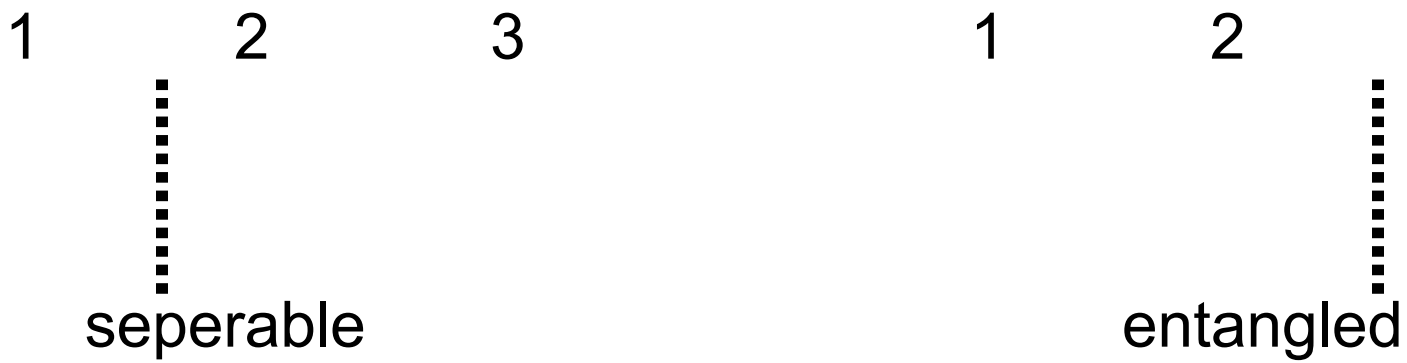
$$|\psi\rangle \otimes \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$$

If we consider qubit 1 as one subsystem and qubits 2 and 3 as another subsystem, then the state is separable across this divide

However, if we consider qubits 1 and 2 as one system and qubits 3 as one subsystem, then the state is entangled across this divide.

$$(\alpha|0\rangle+\beta|1\rangle)\otimes\frac{1}{\sqrt{2}}(|00\rangle+|11\rangle) = \frac{\alpha}{\sqrt{2}}|000\rangle+\frac{\alpha}{\sqrt{2}}|011\rangle+\frac{\beta}{\sqrt{2}}|100\rangle+\frac{\beta}{\sqrt{2}}|111\rangle$$

$$\neq (a_{00}|00\rangle+a_{01}|01\rangle+a_{10}|10\rangle+a_{11}|11\rangle)\otimes(b_0|0\rangle+b_1|1\rangle)$$



Separable, Entangled, 3 Qubits

Sometimes we will deal with entangled states across non adjacent qubits:

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$$

$$\frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$$

How do we even "write" this? $|\psi\rangle_2 \otimes \frac{1}{\sqrt{2}}(|00\rangle_{13} + |11\rangle_{13})$

Subscript denotes which qubit(s) you are talking about.

Separable, Entangled, 3 Qubits

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$$

1 2 3

$$|\psi\rangle_1 \otimes \frac{1}{\sqrt{2}}(|00\rangle_{23} + |11\rangle_{23})$$

$$\frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$$

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$$

1 2 3

$$|\psi\rangle_2 \otimes \frac{1}{\sqrt{2}}(|00\rangle_{13} + |11\rangle_{13})$$

$$\frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$$

Separable, Entangled, 3 Qubits

$$(\alpha|0\rangle_1 + \beta|1\rangle_1 \otimes \frac{1}{\sqrt{2}}(|00\rangle_{23} + |11\rangle_{23})$$

$$= \frac{\alpha}{\sqrt{2}}|000\rangle_{123} + \frac{\alpha}{\sqrt{2}}|011\rangle_{123} + \frac{\beta}{\sqrt{2}}|100\rangle_{123} + \frac{\beta}{\sqrt{2}}|111\rangle_{123}$$

$$= \frac{\alpha}{\sqrt{2}}|000\rangle + \frac{\alpha}{\sqrt{2}}|011\rangle + \frac{\beta}{\sqrt{2}}|100\rangle + \frac{\beta}{\sqrt{2}}|111\rangle$$

$$(\alpha|0\rangle_2 + \beta|1\rangle_2) \otimes \frac{1}{\sqrt{2}}(|00\rangle_{13} + |11\rangle_{13})$$

$$= \frac{\alpha}{\sqrt{2}}|000\rangle_{213} + \frac{\alpha}{\sqrt{2}}|011\rangle_{213} + \frac{\beta}{\sqrt{2}}|100\rangle_{213} + \frac{\beta}{\sqrt{2}}|111\rangle_{213}$$

$$= \frac{\alpha}{\sqrt{2}}|000\rangle + \frac{\alpha}{\sqrt{2}}|101\rangle + \frac{\beta}{\sqrt{2}}|010\rangle + \frac{\beta}{\sqrt{2}}|111\rangle$$

When we don't write subscripts we mean "standard ordering"

Teleportation



ALICE

$$\frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$$



BOB

Alice does not know the wave function

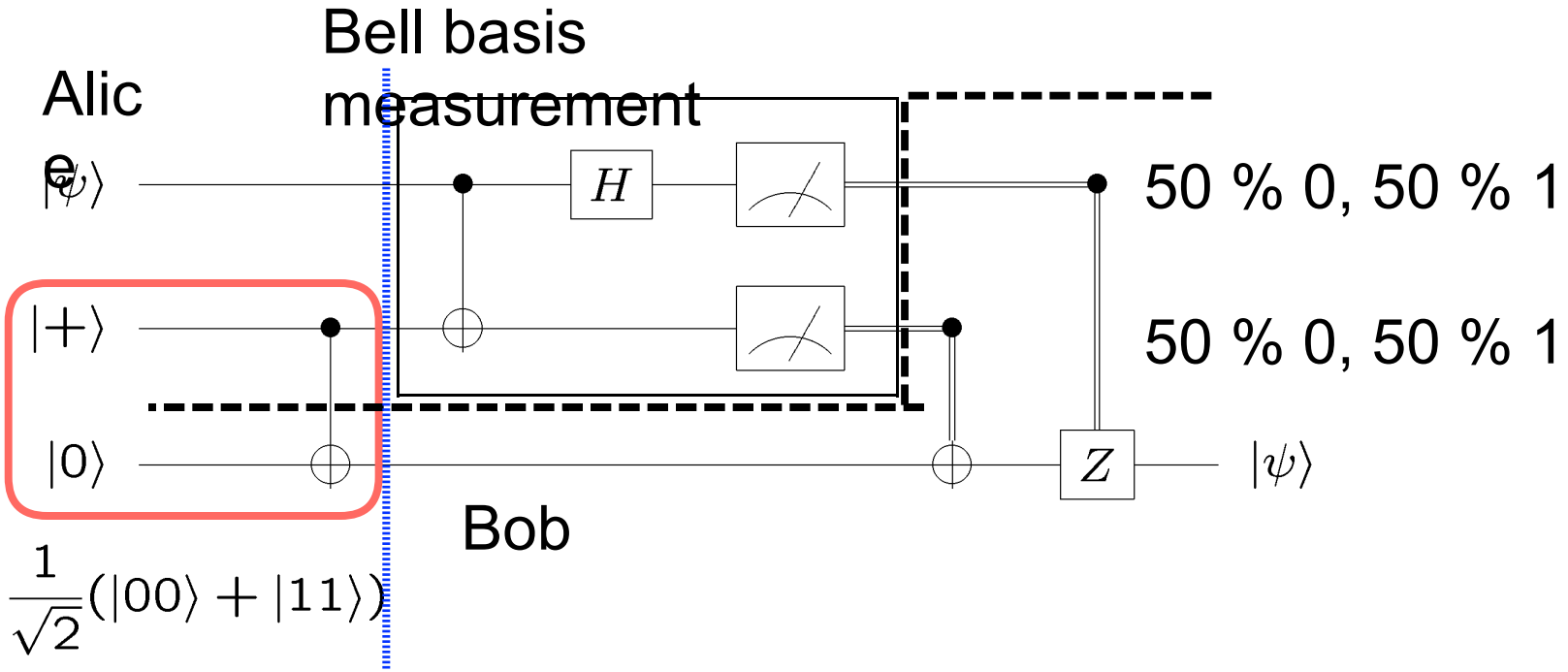
$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$$

We have three qubits whose wave function is

$$|\psi\rangle \otimes \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$$

qubit 1 — qubit 2 and qubit 3

Teleportation



$$|\psi\rangle_1 \otimes \frac{1}{\sqrt{2}}(|00\rangle_{23} + |11\rangle_{23})$$

$$= \frac{\alpha}{\sqrt{2}}|000\rangle + \frac{\alpha}{\sqrt{2}}|011\rangle + \frac{\beta}{\sqrt{2}}|100\rangle + \frac{\beta}{\sqrt{2}}|111\rangle$$

Teleportation

$$|\psi\rangle_1 \otimes \frac{1}{\sqrt{2}}(|00\rangle_{23} + |11\rangle_{23})$$

$$= \frac{\alpha}{\sqrt{2}}|000\rangle + \frac{\alpha}{\sqrt{2}}|011\rangle + \frac{\beta}{\sqrt{2}}|100\rangle + \frac{\beta}{\sqrt{2}}|111\rangle$$

Express this state in terms of Bell basis for first two qubits.

Bell basis

$$|\Phi_+\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$$

$$|\Phi_-\rangle = \frac{1}{\sqrt{2}}(|00\rangle - |11\rangle)$$

$$|\Psi_+\rangle = \frac{1}{\sqrt{2}}(|01\rangle + |10\rangle)$$

$$|\Psi_-\rangle = \frac{1}{\sqrt{2}}(|01\rangle - |10\rangle)$$

Computational basis

$$|00\rangle = \frac{1}{\sqrt{2}}(|\Phi_+\rangle + |\Phi_-\rangle)$$

$$|01\rangle = \frac{1}{\sqrt{2}}(|\Psi_+\rangle + |\Psi_-\rangle)$$

$$|10\rangle = \frac{1}{\sqrt{2}}(|\Phi_+\rangle - |\Phi_-\rangle)$$

$$|01\rangle = \frac{1}{\sqrt{2}}(|\Phi_+\rangle - |\Phi_-\rangle)$$

Teleportation

Bell basis

$$|\Phi_+\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$$
$$|\Phi_-\rangle = \frac{1}{\sqrt{2}}(|00\rangle - |11\rangle)$$
$$|\Psi_+\rangle = \frac{1}{\sqrt{2}}(|01\rangle + |10\rangle)$$
$$|\Psi_-\rangle = \frac{1}{\sqrt{2}}(|01\rangle - |10\rangle)$$

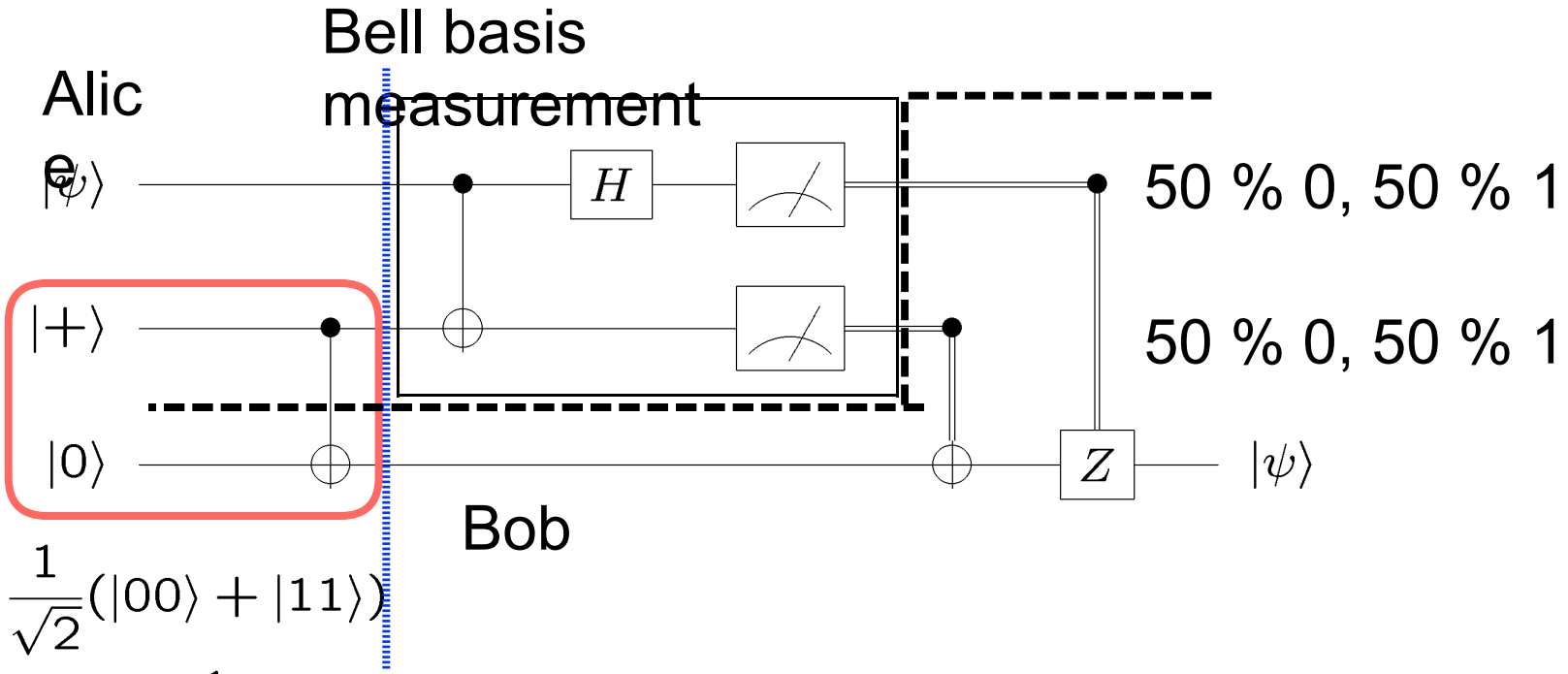
Computational basis

$$|00\rangle = \frac{1}{\sqrt{2}}(|\Phi_+\rangle + |\Phi_-\rangle)$$
$$|01\rangle = \frac{1}{\sqrt{2}}(|\Psi_+\rangle + |\Psi_-\rangle)$$
$$|11\rangle = \frac{1}{\sqrt{2}}(|\Phi_+\rangle - |\Phi_-\rangle)$$
$$|10\rangle = \frac{1}{\sqrt{2}}(|\Psi_+\rangle - |\Psi_-\rangle)$$

$$\frac{\alpha}{\sqrt{2}}|000\rangle + \frac{\alpha}{\sqrt{2}}|011\rangle + \frac{\beta}{\sqrt{2}}|100\rangle + \frac{\beta}{\sqrt{2}}|111\rangle$$

$$\begin{aligned} &= \frac{1}{2}(\alpha(|\Phi_+\rangle + |\Phi_-\rangle) \otimes |0\rangle + \alpha(|\Psi_+\rangle + |\Psi_-\rangle) \otimes |1\rangle \\ &\quad +\beta(|\Psi_+\rangle - |\Psi_-\rangle) \otimes |0\rangle + \beta(|\Phi_+\rangle - |\Phi_-\rangle) \otimes |1\rangle \\ &= \frac{1}{2}\Big[|\Phi_+\rangle \otimes (\alpha|0\rangle + \beta|1\rangle) + |\Phi_-\rangle \otimes (\alpha|0\rangle - \beta|1\rangle) \\ &\quad +|\Psi_+\rangle \otimes (\alpha|1\rangle + \beta|0\rangle) + |\Psi_-\rangle \otimes (\alpha|1\rangle - \beta|0\rangle)\Big] \end{aligned}$$

Teleportation



$$|\psi\rangle_1 \otimes \frac{1}{\sqrt{2}}(|00\rangle_{23} + |11\rangle_{23})$$

$$= \frac{\alpha}{\sqrt{2}}|000\rangle + \frac{\alpha}{\sqrt{2}}|011\rangle + \frac{\beta}{\sqrt{2}}|100\rangle + \frac{\beta}{\sqrt{2}}|111\rangle$$

$$= \frac{1}{2}\Big[|\Phi_+\rangle_{12}(\alpha|0\rangle_3 + \beta|1\rangle_3) + |\Phi_-\rangle_{12}(\alpha|0\rangle_3 - \beta|1\rangle_3)$$

$$+|\Psi_+\rangle_{12}(\alpha|1\rangle_3 + \beta|0\rangle_3) + |\Psi_-\rangle_{12}(\alpha|1\rangle_3 - \beta|0\rangle_3)\Big]$$

Dropping The Tensor Symbol

$$\frac{1}{2}\Big[|\Phi_+\rangle \otimes (\alpha|0\rangle + \beta|1\rangle + |\Phi_-\rangle \otimes (\alpha|0\rangle - \beta|1\rangle)$$
$$+|\Psi_+\rangle \otimes (\alpha|1\rangle + \beta|0\rangle) + |\Psi_-\rangle \otimes (\alpha|1\rangle - \beta|0\rangle)\Big]$$

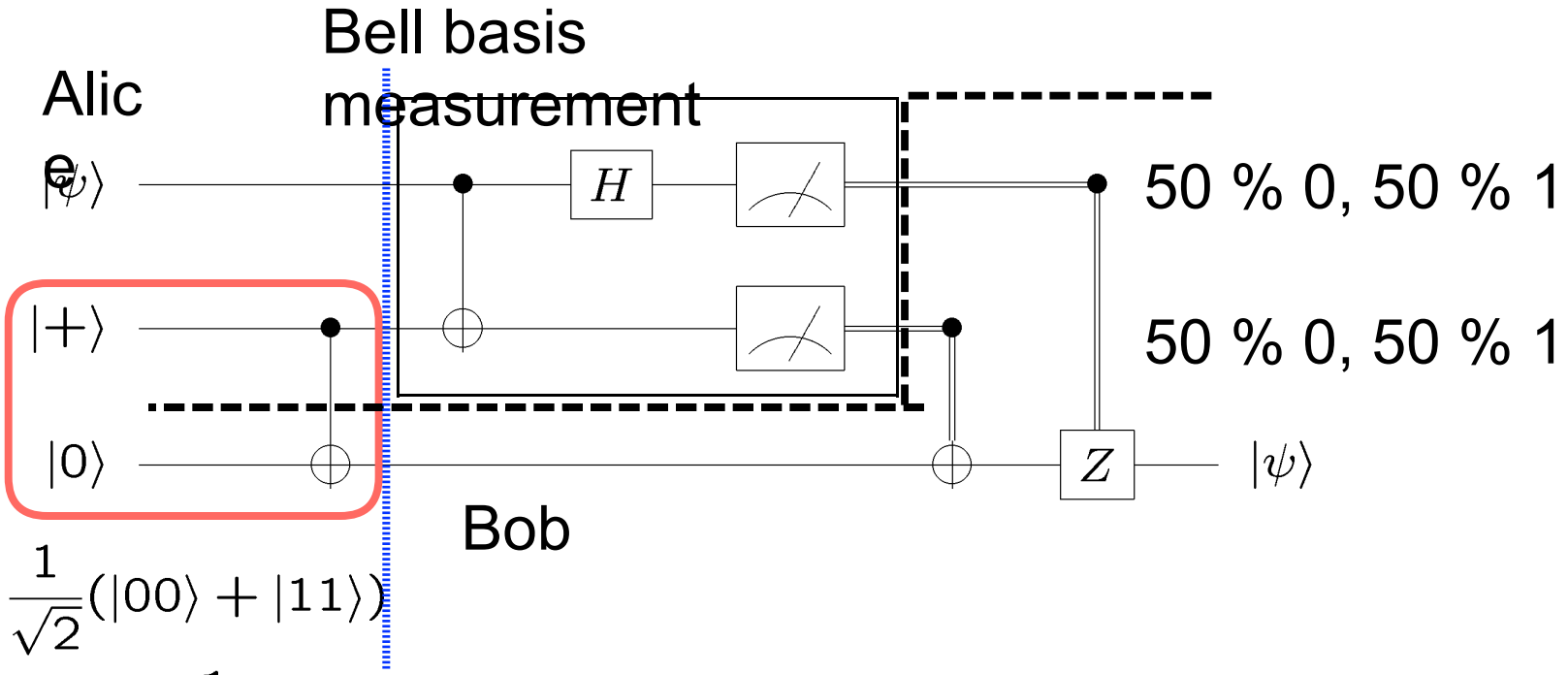
$$= \frac{1}{2}\Big[|\Phi_+\rangle_{12}(\alpha|0\rangle_3 + \beta|1\rangle_3) + |\Phi_-\rangle_{12}(\alpha|0\rangle_3 - \beta|1\rangle_3)$$
$$+|\Psi_+\rangle_{12}(\alpha|1\rangle_3 + \beta|0\rangle_3) + |\Psi_-\rangle_{12}(\alpha|1\rangle_3 - \beta|0\rangle_3)\Big]$$

$$= \frac{1}{2}\Big[|\Phi_+\rangle(\alpha|0\rangle) + \beta|1\rangle) + |\Phi_-\rangle(\alpha|0\rangle - \beta|1\rangle)$$
$$+|\Psi_+\rangle(\alpha|1\rangle + \beta|0\rangle) + |\Psi_-\rangle(\alpha|1\rangle - \beta|0\rangle)\Big]$$

Sometimes we will just “drop” the tensor symbol.

“Context” lets us know that there is an implicit tensor product.

Teleportation



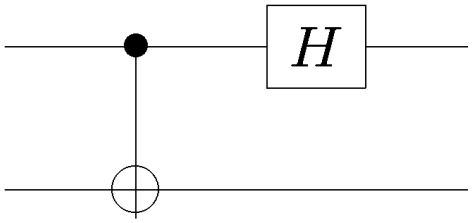
$$|\psi\rangle_1 \otimes \frac{1}{\sqrt{2}}(|00\rangle_{23} + |11\rangle_{23})$$

$$= \frac{\alpha}{\sqrt{2}}|000\rangle + \frac{\alpha}{\sqrt{2}}|011\rangle + \frac{\beta}{\sqrt{2}}|100\rangle + \frac{\beta}{\sqrt{2}}|111\rangle$$

$$= \frac{1}{2}\Big[|\Phi_+\rangle_{12}(\alpha|0\rangle_3) + \beta|1\rangle_3) + |\Phi_-\rangle_{12}(\alpha|0\rangle_3 - \beta|1\rangle_3)$$

$$+|\Psi_+\rangle_{12}(\alpha|1\rangle_3 + \beta|0\rangle_3) + |\Psi_-\rangle_{12}(\alpha|1\rangle_3 - \beta|0\rangle_3)\Big]$$

Bell Basis Measurement



$$|\Phi_+\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle) \rightarrow |00\rangle \quad |\Phi_-\rangle = \frac{1}{\sqrt{2}}(|00\rangle - |11\rangle) \rightarrow |10\rangle$$

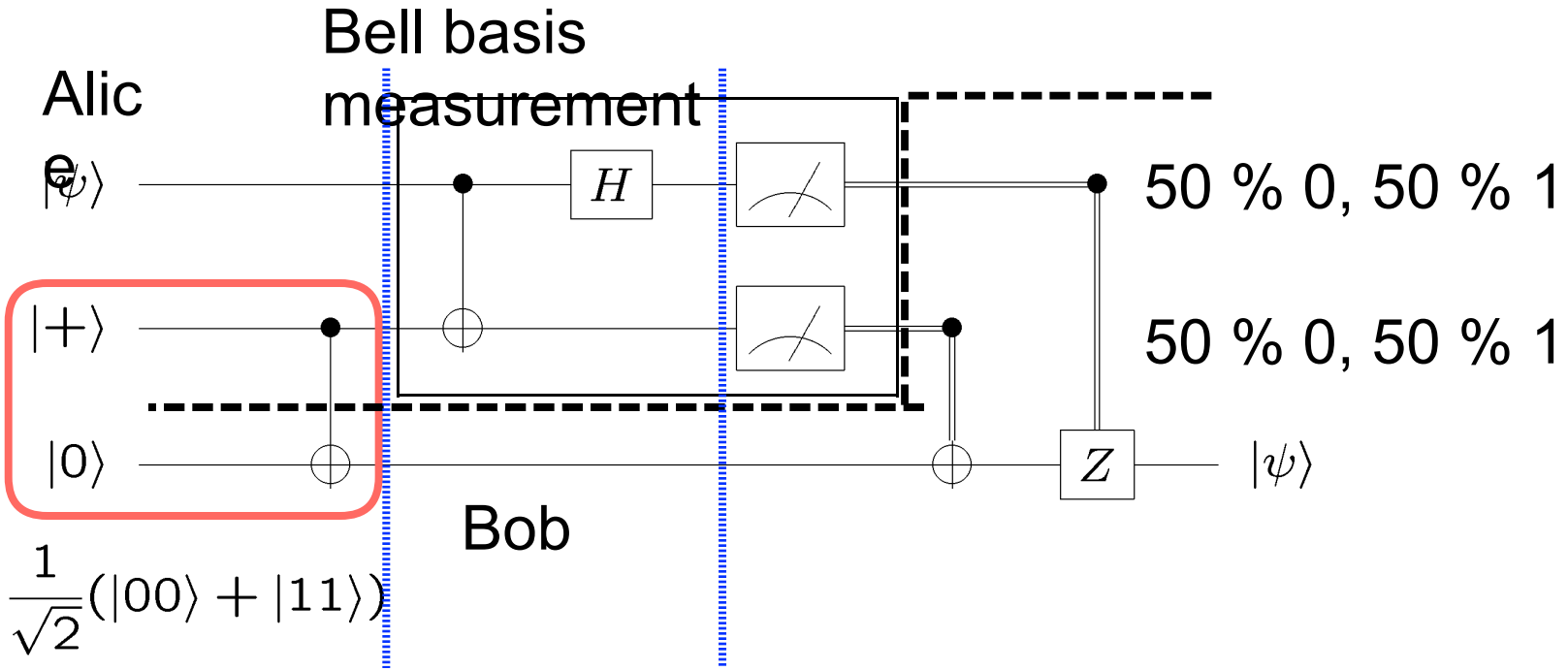
$$|\Psi_+\rangle = \frac{1}{\sqrt{2}}(|01\rangle + |10\rangle) \rightarrow |01\rangle \quad |\Psi_-\rangle = \frac{1}{\sqrt{2}}(|01\rangle - |10\rangle) \rightarrow |11\rangle$$

$$\frac{1}{2}\Big[|\Phi_+\rangle_{12}(\alpha|0\rangle_3 + \beta|1\rangle_3) + |\Phi_-\rangle_{12}(\alpha|0\rangle_3 - \beta|1\rangle_3)$$
$$+|\Psi_+\rangle_{12}(\alpha|1\rangle_3 + \beta|0\rangle_3) + |\Psi_-\rangle_{12}(\alpha|1\rangle_3 - \beta|0\rangle_3)\Big]$$

$$\downarrow$$

$$\frac{1}{2}[|00\rangle_{12}(\alpha|0\rangle_3 + \beta|1\rangle_3) + |10\rangle_{12}(\alpha|0\rangle_3 - \beta|1\rangle_3)$$
$$+|01\rangle_{12}(\alpha|1\rangle_3 + \beta|0\rangle_3) + |11\rangle_{12}(\alpha|1\rangle_3 - \beta|0\rangle_3)]$$

Teleportation



$$\frac{1}{2}\Big[|\Phi_+\rangle_{12}(\alpha|0\rangle_3 + \beta|1\rangle_3) + |\Phi_-\rangle_{12}(\alpha|0\rangle_3 - \beta|1\rangle_3)$$
$$+|\Psi_+\rangle_{12}(\alpha|1\rangle_3 + \beta|0\rangle_3) + |\Psi_-\rangle_{12}(\alpha|1\rangle_3 - \beta|0\rangle_3)\Big]$$

$$\frac{1}{2}[|00\rangle_{12}(\alpha|0\rangle_3 + \beta|1\rangle_3) + |10\rangle_{12}(\alpha|0\rangle_3 - \beta|1\rangle_3)$$
$$+|01\rangle_{12}(\alpha|1\rangle_3 + \beta|0\rangle_3) + |11\rangle_{12}(\alpha|1\rangle_3 - \beta|0\rangle_3)]$$

Teleportation

Given the wave fun

$$\frac{1}{2}[|00\rangle_{12}(\alpha|0\rangle_3+\beta|1\rangle_3)+|10\rangle_{12}(\alpha|0\rangle_3-\beta|1\rangle_3)$$
$$+|01\rangle_{12}(\alpha|1\rangle_3+\beta|0\rangle_3)+|11\rangle_{12}(\alpha|1\rangle_3-\beta|0\rangle_3)]$$

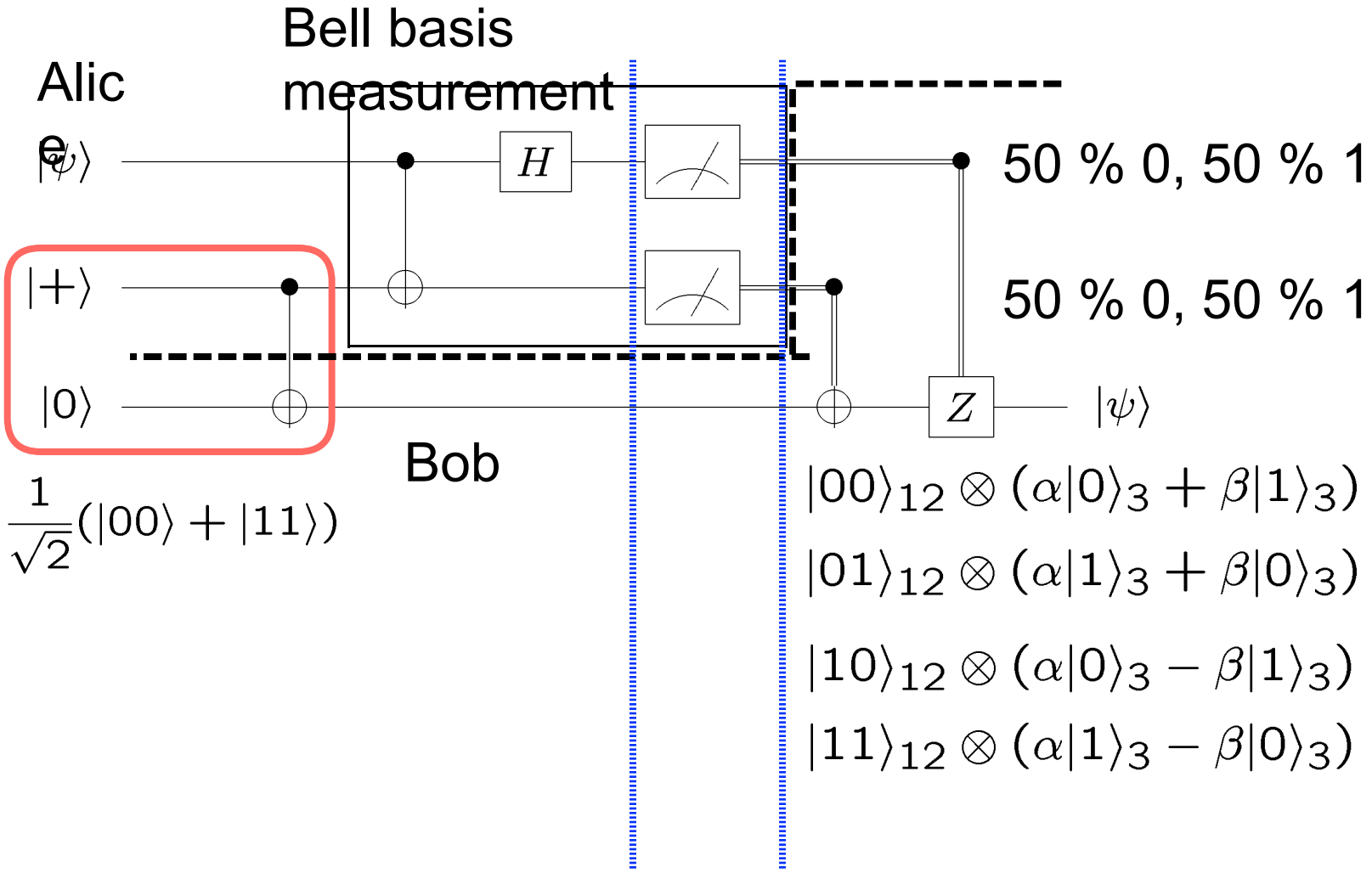
Measure the first two qubits in the computational basis

$$M_{00}=|00\rangle\langle 00|\otimes I \qquad M_{01}=|01\rangle\langle 01|\otimes I$$
$$M_{10}=|10\rangle\langle 10|\otimes I \qquad M_{11}=|11\rangle\langle 11|\otimes I$$

Equal ¼ probability for all four outcomes and new states are:

$$|00\rangle_{12}\otimes(\alpha|0\rangle_3+\beta|1\rangle_3)$$
$$|01\rangle_{12}\otimes(\alpha|1\rangle_3+\beta|0\rangle_3)$$
$$|10\rangle_{12}\otimes(\alpha|0\rangle_3-\beta|1\rangle_3)$$
$$|11\rangle_{12}\otimes(\alpha|1\rangle_3-\beta|0\rangle_3)$$

Teleportation



$$\frac{1}{2}[|00\rangle_{12}(\alpha|0\rangle_3 + \beta|1\rangle_3) + |10\rangle_{12}(\alpha|0\rangle_3 - \beta|1\rangle_3)$$
$$+|01\rangle_{12}(\alpha|1\rangle_3 + \beta|0\rangle_3) + |11\rangle_{12}(\alpha|1\rangle_3 - \beta|0\rangle_3)]$$

Teleportation

If the bits sent from Alice to Bob are 00, do nothing

$$|00\rangle_{12} \otimes (\alpha|0\rangle_3 + \beta|1\rangle_3) = |00\rangle_{12} \otimes |\psi\rangle_3$$

If the bits sent from Alice to Bob are 01, apply a bit flip

$$(I_4 \otimes X)|01\rangle_{12} \otimes (\alpha|1\rangle_3 + \beta|0\rangle_3) = |01\rangle_{12} \otimes (\alpha|0\rangle_3 + \beta_1|1\rangle_3)$$

$$= |01\rangle_{12} \otimes |\psi\rangle_3$$

If the bits sent from Alice to Bob are 10, apply a phase flip

$$(I_4 \otimes Z)|10\rangle_{12} \otimes (\alpha|0\rangle_3 - \beta|1\rangle_3) = |10\rangle \otimes (\alpha|0\rangle_3 + \beta|1\rangle_3)$$

$$= |10\rangle_{12} \otimes |\psi\rangle_3$$

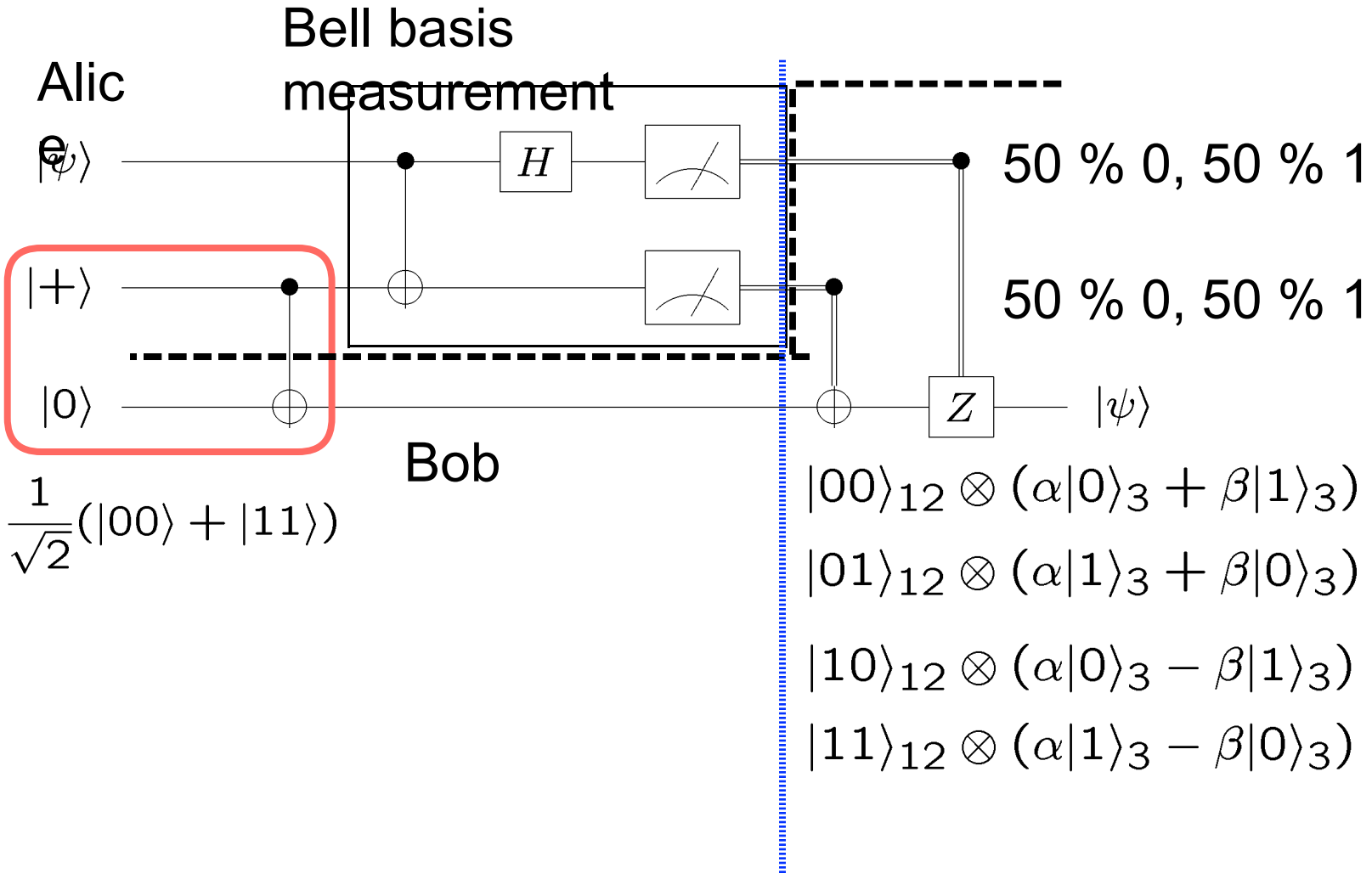
If the bits sent from Alice to Bob are 11, apply a bit & phase flip

$$(I_4 \otimes Z)(I_4 \otimes X)|11\rangle_{12} \otimes (\alpha|1\rangle_3 - \beta|0\rangle_3) = (I_4 \otimes Z)|11\rangle_{12} \otimes (\alpha|0\rangle_3 - \beta|1\rangle_3)$$

$$= |11\rangle_{12} \otimes (\alpha|0\rangle_3 + \beta|1\rangle_3)$$

$$= |11\rangle_{12} \otimes |\psi\rangle_3$$

Teleportation



Teleportation



Alice



Bob

$\cong$

$|\psi\rangle$

Pauli

2 bits

Bell measurement

$\frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$



$|\psi\rangle$

Alice



Bob

Teleportation

1 qubit = 1 ebit + 2 bits

Teleportation says we can replace transmitting a qubit with a shared entangled pair of qubits plus two bits of classical communication.

Superdense Coding

Next we will see that

2 bits = 1 qubit + 1 ebit

Superdense Coding

Suppose Alice and Bob each have one qubit and the joint two qubit wave function is the entangled state

$$|\Phi_+\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$$

Alice wants to send two bits to Bob. Call these bits b_1 and b_2

Alice applies the following operator to her qubit: $Z^{b_2}X^{b_1}$

Alice then sends her qubit to Bob.

note: $M^0 = I$

Bob then measures in the Bell basis to determine the two bits

2 bits = 1 qubit + 1 ebit

In Class Problem 1

Bell Basis

The four Bell states can be turned into each other using operations on only one of the qubits:

$$|\Phi_+\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$$

$$(X\otimes I)|\Phi_+\rangle = (X\otimes I)\frac{1}{\sqrt{2}}(|00\rangle+|11\rangle) = \frac{1}{\sqrt{2}}(|10\rangle+|01\rangle) = |\Psi_+\rangle$$

$$(Z\otimes I)|\Phi_+\rangle = (Z\otimes I)\frac{1}{\sqrt{2}}(|00\rangle+|11\rangle) = \frac{1}{\sqrt{2}}(|00\rangle-|11\rangle) = |\Phi_-\rangle$$

$$(ZX\otimes I)|\Phi_+\rangle = (ZX\otimes I)\frac{1}{\sqrt{2}}(|00\rangle+|11\rangle) = \frac{1}{\sqrt{2}}(-|10\rangle+|01\rangle) = |\Psi_-\rangle$$

Superdense Coding

Initially $|\Phi_+\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$

:

Alice applies the following operator to her qubit: $Z^{b_2}X^{b_1}$

$$(Z^{b_2}X^{b_1} \otimes I)|\Phi_+\rangle$$

$$\begin{aligned}
b_1 = 0, b_2 = 0 &\qquad |\Phi_+\rangle \\
b_1 = 0, b_2 = 1 &\qquad (Z \otimes I)|\Phi_+\rangle = |\Phi_-\rangle \\
b_1 = 1, b_2 = 0 &\qquad (X \otimes I)|\Phi_+\rangle = |\Psi_+\rangle \\
b_1 = 1, b_2 = 1 &\qquad (ZX \otimes I)|\Phi_+\rangle = |\Psi_-\rangle
\end{aligned}$$

Bob can uniquely determine which of the four states he has and thus figure out Alice's two bits!

Superdense Coding



b_1

b_2

$Z^{b_2} X^{b_1}$

$|\Phi_+\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$



b_1

b_2

Bell basis measurement

Teleportation

1 qubit = 1 ebit + 2 bits

Teleportation says we can replace transmitting a qubit with a shared entangled pair of qubits plus two bits of classical communication.

Superdense Coding

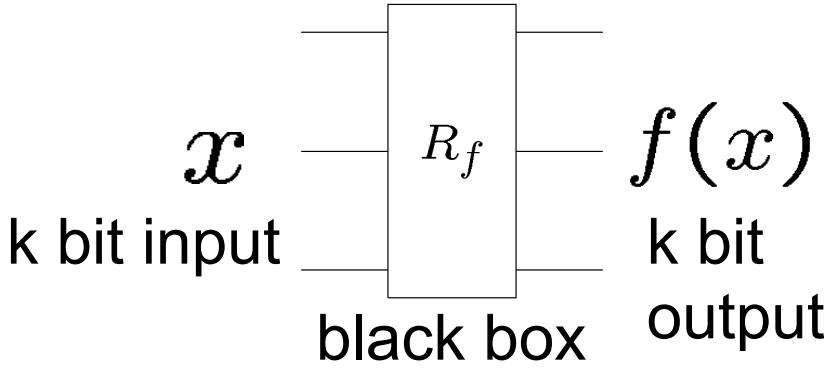
2 bits = 1 qubit + 1 ebit

We can send two bits of classical information if we share an entangled state and can communicate one qubit of quantum information:

Quantum Algorithms

Classical Promise Problem Query Complexity

Given: A black box which computes some function



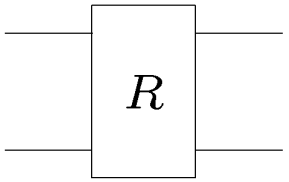
Promise: the function belongs to a set $\mathcal{S}$ which is a subset of all possible functions.

Properties: the set $\mathcal{S}$ can be divided into disjoint subsets $\mathcal{S}_1, \mathcal{S}_2, \ldots, \mathcal{S}_m$

Problem: What is the minimal number of times we have to use (query) the black box in order to determine which subset $\mathcal{S}_i$ the function belongs to?

Example

Suppose you are given a black box which computes one of the following four reversible classical gates:



2 bits input

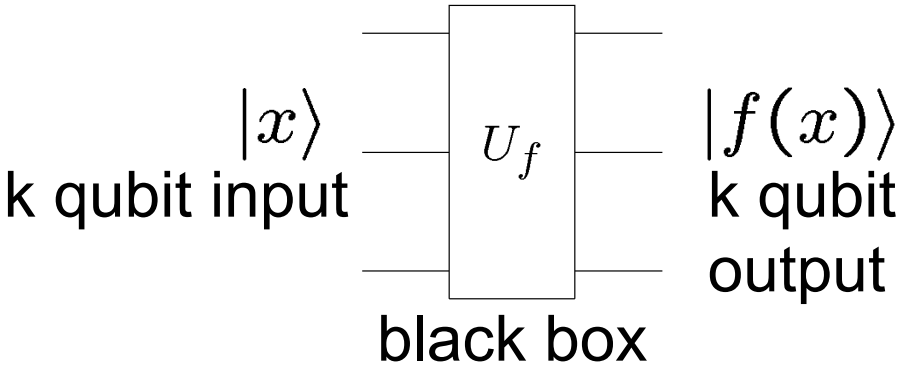
2 bits output

$00 \to 00$	$00 \to 01$	$00 \to 00$	$00 \to 01$
$01 \to 01$	$01 \to 00$	$01 \to 01$	$01 \to 00$
$10 \to 10$	$10 \to 11$	$10 \to 11$	$10 \to 10$
$11 \to 11$	$11 \to 10$	$11 \to 10$	$11 \to 11$
"identity"	NOT 2nd bit	controlled-NOT	controlled-NOT + NOT 2nd bit
	$\mathcal{S}_1$		$\mathcal{S}_2$

Deutsch's (Classical) Problem: What is the minimal number of times we have to use this black box to determine whether we are given one of the first two or the second two functions?

Quantum Promise Query Complexity

Given: A quantum gate which, when used as a classical device computes a reversible function



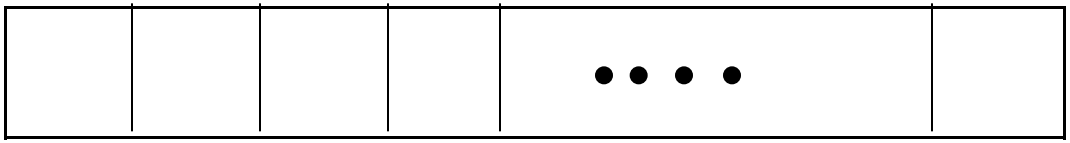
Promise: the function belongs to a set $\mathcal{S}$ which is a subset of all possible functions.

Properties: the set $\mathcal{S}$ can be divided into disjoint subsets $\mathcal{S}_1, \mathcal{S}_2, \ldots, \mathcal{S}_m$

Problem: What is the minimal number of times we have to use (query) the quantum gate in order to determine which subset $\mathcal{S}_i$ the function belongs to?

n Qubit Registers

Up until now, we have dealt with only 1,2,3, or 4 qubits.
Now we will deal with n qubits at a time!



n
qubits

Computational basis:

$$|x_1\rangle \otimes |x_2\rangle \otimes |x_3\rangle \otimes \cdots \otimes |x_n\rangle \qquad x_i \in \{0,1\}$$

$$|x_1, x_2, x_3, \ldots, x_n\rangle$$

$|x\rangle \quad x \in \{0,1\}^n$ n bit string

$$|x\rangle \quad x \in \{0,1,2,\ldots,2^n-1\}$$

n Qubit States

n qubits have a wave function with 2^n complex numbers.

Writing 2^n complex numbers down, and keeping track of them (in a naïve manner) is very computationally inefficient.

This is one of the first indications that simulating a quantum computer on a classical computer might be very difficult.

$$|\psi\rangle = \sum_{x=0}^{2^n-1} a_x |x\rangle$$

a_x are 2^n complex numbers

properly normalized: $\sum_{x=0}^{2^n-1} |a_x|^2 = 1$

n Qubit States

Example:

$$|\psi\rangle = \sum_{x=0}^{2^n-1} a_x |x\rangle \quad a_x = \frac{1}{\sqrt{2^n}}$$

properly normalized: $\sum_{x=0}^{2^n-1} |a_x|^2 = \sum_{x=0}^{2^n-1} \frac{1}{2^n} = 1$

$n = 3$

$$|\psi\rangle = \sum_{x=0}^{7} \frac{1}{2\sqrt{2}} |x\rangle$$

$$= \frac{1}{2\sqrt{2}}(|0\rangle + |1\rangle + |2\rangle + |3\rangle + |4\rangle + |5\rangle + |6\rangle + |7\rangle)$$

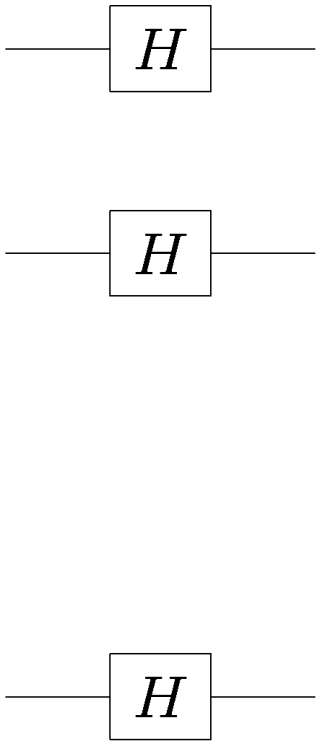
$$= \frac{1}{2\sqrt{2}}(|000\rangle + |001\rangle + |010\rangle + |011\rangle + |100\rangle + |101\rangle + |110\rangle + |111\rangle)$$

Notice how compact this 1st notation

n Qubit Hadamard

Hadamard all n qubits $H^{\otimes n} = H \otimes H \otimes \cdots \otimes H$

$$H = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$$



n qubits input

n qubits output

n Qubit Hadamard

Hadamard one qubit in computational basis:

$$H|x_i\rangle = \frac{1}{\sqrt{2}}(|0\rangle + (-1)^{x_i}|1\rangle) = \frac{1}{\sqrt{2}} \sum_{y_i=0,1} (-1)^{x_i y_i}|y_i\rangle$$

Hadamard n qubits in computational basis:

$$H^{\otimes n}|x\rangle = (H \otimes H \otimes \cdots \otimes H)|x_1\rangle \otimes |x_2\rangle \otimes \cdots \otimes |x_n\rangle$$

$$= (H|x_1\rangle) \otimes (H|x_2\rangle) \otimes \cdots \otimes (H|x_n\rangle)$$

$$= \frac{1}{\sqrt{2}}(|0\rangle + (-1)^{x_1}|1\rangle) \otimes \frac{1}{\sqrt{2}}(|0\rangle + (-1)^{x_2}|1\rangle) \otimes \cdots \otimes \frac{1}{\sqrt{2}}(|0\rangle + (-1)^{x_n}|1\rangle)$$

$$= \frac{1}{\sqrt{2^n}} \sum_{y_1,y_2,\ldots,y_n=0}^{1} (-1)^{y_1x_1 + y_2x_2 + \cdots y_nx_n}|y_1, y_2, \ldots, y_n\rangle$$

$$y \cdot x = y_1x_1 + y_2x_2 + \cdots + y_nx_n$$

$$= \frac{1}{\sqrt{2^n}} \sum_{y=0}^{2^n-1} (-1)^{y \cdot x}|y\rangle$$

n Qubit Hadamard

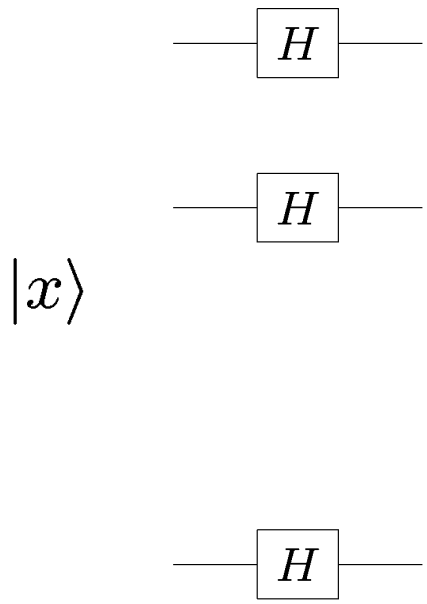
$$H^{\otimes n}|x\rangle = \frac{1}{\sqrt{2^n}} \sum_{y=0}^{2^n-1} (-1)^{y \cdot x} |y\rangle$$

$$y \cdot x = y_1 x_1 + y_2 x_2 + \cdots + y_n x_n$$

Addition can be done modulo 2 (turns plus to exclusive-or)

$$y \cdot x = y_1 x_1 \oplus y_2 x_2 \oplus \cdots \oplus y_n x_n$$

$$b_1 \oplus b_2 = \begin{cases} 0 \text{ if } b_1 = b_2 \\ 1 \text{ if } b_1 \neq b_2 \end{cases}$$



$$\frac{1}{\sqrt{2^n}} \sum_{y=0}^{2^n-1} (-1)^{y \cdot x} |y\rangle$$

Again notice compactness of notation.

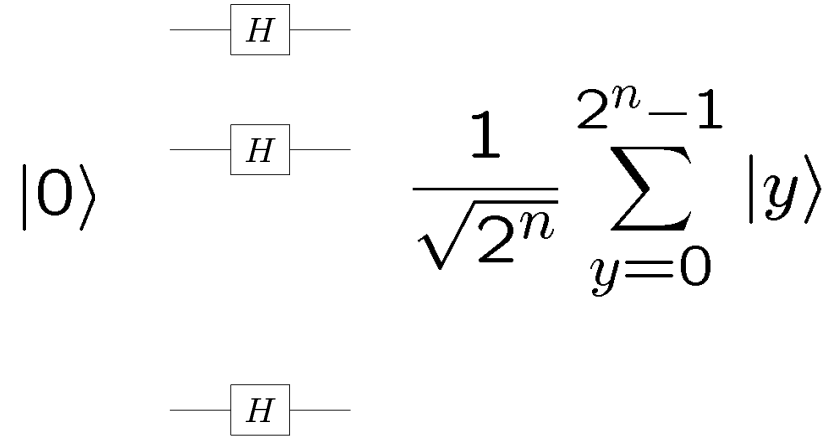
Superposition Over All

$$H^{\otimes n}|x\rangle = \frac{1}{\sqrt{2^n}} \sum_{y=0}^{2^n-1} (-1)^{y \cdot x} |y\rangle$$

$$y \cdot x = y_1 x_1 \oplus y_2 x_2 \oplus \cdots \oplus y_n x_n$$

If we start in the zero bitstring, then Hadmarding all n qubits creates a superposition over all possible bitstrings:

$$H^{\otimes n}|0\rangle = \frac{1}{\sqrt{2^n}} \sum_{y=0}^{2^n-1} |y\rangle$$



$$|0\rangle \qquad \frac{1}{\sqrt{2^n}} \sum_{y=0}^{2^n-1} |y\rangle$$

Superposition Over All

Hadamarding the superposition over all states:

$$H^{\otimes n}\left(\frac{1}{\sqrt{2^n}}\sum_{y=0}^{2^n-1}|y\rangle\right) = \frac{1}{\sqrt{2^n}}\sum_{y=0}^{2^n-1}H^{\otimes n}|y\rangle$$

$$= \frac{1}{\sqrt{2^n}}\sum_{y=0}^{2^n-1}\frac{1}{\sqrt{2^n}}\sum_{z=0}^{2^n-1}(-1)^{z\cdot y}|z\rangle$$

$$= \frac{1}{2^n}\sum_{z=0}^{2^n-1}\left(\sum_{y=0}^{2^n-1}(-1)^{z\cdot y}\right)|z\rangle$$

$$\frac{1}{2^n}\sum_{y=0}^{2^n-1}(-1)^{z\cdot y} = \frac{1}{2^n}\sum_{y_1,\ldots,y_n=0}^{1}(-1)^{z_1y_1+\cdots+z_ny_n}$$

Superposition Over All

$$\frac{1}{2^n}\sum_{y=0}^{2^n-1}(-1)^{z\cdot y} = \frac{1}{2^n}\sum_{y_1,\ldots,y_n=0}^{1}(-1)^{z_1y_1+\cdots+z_ny_n}$$

$$= \frac{1}{2^n}\sum_{y_1,\ldots,y_n=0}^{1}(-1)^{z_1y_1}(-1)^{z_2y_2}\cdots(-1)^{z_ny_n}$$

$$= \left[\frac{1}{2}\sum_{y_1=0}^{1}(-1)^{z_1y_1}\right]\left[\frac{1}{2}\sum_{y_2=0}^{1}(-1)^{z_2y_2}\right]\cdots\left[\frac{1}{2}\sum_{y_n=0}^{1}(-1)^{z_ny_n}\right]$$

$$\frac{1}{2}\sum_{y_i=0}^{1}(-1)^{z_iy_i} = \frac{1}{2}((-1)^0+(-1)^{z_i}) = \delta_{z_i,0}$$

$$\frac{1}{2^n}\sum_{y=0}^{2^n-1}(-1)^{z\cdot y} = \delta_{z_1,0}\delta_{z_2,0}\cdots\delta_{z_n,0} = \delta_{z,0}$$

Superposition Over All

$$H^{\otimes n}\left(\frac{1}{\sqrt{2^n}}\sum_{y=0}^{2^n-1}|y\rangle\right)=\frac{1}{2^n}\sum_{z=0}^{2^n-1}\left(\sum_{y=0}^{2^n-1}(-1)^{z\cdot y}\right)|z\rangle$$

$$\frac{1}{2^n}\sum_{y=0}^{2^n-1}(-1)^{z\cdot y}=\delta_{z_1,0}\delta_{z_2,0}\cdots\delta_{z_n,0}=\delta_{z,0}$$

$$H^{\otimes n}\left(\frac{1}{\sqrt{2^n}}\sum_{y=0}^{2^n-1}|y\rangle\right)=\sum_{z=0}^{2^n-1}\delta_{z,0}|z\rangle=|0\rangle$$



Could have found in easier fashion using



$$(H^{\otimes n})(H^{\otimes n})=(H^2)^{\otimes n}=I$$



From Comp. Basis to Matrix

From the effect of the Hadamard on the computational basis

$$H^{\otimes n}|x\rangle = \frac{1}{\sqrt{2^n}} \sum_{y=0}^{2^n-1} (-1)^{y \cdot x} |y\rangle$$

We can deduce the form of the matrix in outer product form:

$$H^{\otimes n} = \frac{1}{\sqrt{2^n}} \sum_{x,y=0}^{2^n-1} (-1)^{y \cdot x} |y\rangle\langle x|$$

Hadamard Basis Elements

Recall that the columns of a matrix form a basis.

What is this basis for the Hadamard?

$$H^{\otimes n} = \frac{1}{\sqrt{2^n}} \sum_{x,y=0}^{2^n-1} (-1)^{y \cdot x} |y\rangle\langle x|$$

The basis elements for the Hadmard are:

$$|\phi_x\rangle = \frac{1}{\sqrt{2^n}} \sum_{y=0}^{2^n-1} (-1)^{y \cdot x} |y\rangle$$

Hadamard Basis Elements

Check orthonormality:

$$|\phi_x\rangle = \frac{1}{\sqrt{2^n}} \sum_{y=0}^{2^n-1} (-1)^{y\cdot x} |y\rangle$$

$$\langle \phi_{x'}|\phi_x\rangle = \frac{1}{\sqrt{2^n}} \sum_{y'=0}^{2^n-1} (-1)^{x'\cdot y'} \langle y'| \frac{1}{\sqrt{2^n}} \sum_{y=0}^{2^n-1} (-1)^{y\cdot x} |y\rangle$$

$$\langle \phi_{x'}|\phi_x\rangle = \frac{1}{2^n} \sum_{y,y'=0}^{2^n-1} (-1)^{x'\cdot y' + x\cdot y} \langle y|y'\rangle \longleftarrow \delta_{y,y'}$$

$$\langle \phi_{x'}|\phi_x\rangle = \frac{1}{2^n} \sum_{y=0}^{2^n-1} (-1)^{x'\cdot y + x\cdot y}$$

Hadamard Basis Elements

$$\langle \phi_{x'} | \phi_x \rangle = \frac{1}{2^n} \sum_{y=0}^{2^n-1} (-1)^{x' \cdot y + x \cdot y}$$

$$x' \cdot y + x \cdot y = y_1 x'_1 + y_2 x'_2 + \cdots + y_n x'_n + y_1 x_1 + y_2 x_2 + \cdots + y_n x_n$$

$$= y_1 (x'_1 + x_1) + y_2 (x'_2 + x_2) + \cdots + y_n (x'_n + x_n)$$

$$\langle \phi_{x'} | \phi_x \rangle = \frac{1}{2^n} \sum_{y=0}^{2^n-1} (-1)^{(x' \oplus x) \cdot y}$$

$$x \oplus x' = (x_1 \oplus x'_1, x_2 \oplus x'_2, \ldots, x_n \oplus x'_n)$$

$$\frac{1}{2^n} \sum_{y=0}^{2^n-1} (-1)^{z \cdot y} = \delta_{z_1,0} \delta_{z_2,0} \cdots \delta_{z_n,0} = \delta_{z,0}$$

$$\langle \phi_{x'} | \phi_x \rangle = \delta_{x \oplus x',0} = \delta_{x,x'}$$

In Class Problem 2